JANGAM UMESH CHANDRA

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Education

TEESSIDE UNIVERSITY **Middlesbrough, United Kingdom**
MSc Civil and Structural Engineering. **Sept-2021 - June-2023**
Award – Graduated with Merit
Thesis - Comparison of a residential building in seismic zones in 2 different countries (India & Italy).
The aim of this project is to compare the design of a residential building in two different countries, India and Italy, both located in high seismic zones. Specifically, the project focuses on seismic Zone-5 in India, which includes regions most prone to earthquakes. The project involves identifying the differences between the Indian Standard (IS) codes and the Eurocodes, particularly IS 456:2000 and EN 1991-1-4, in relation to the construction of a residential building. Additionally, a 3D model of the structure will be created using SPACE GASS software, and the bending moment, shear force, and axial force will be calculated for both countries' seismic design requirements.

GITAM DEEMED TO BE UNIVERSITY **Visakhapatnam, India**
B-Tech in Civil Engineering. **June-2017 – June-2021**
Award – Graduated with Distinction; GPA – 8.04/10
Project - Analysis and Design G+6 building for different seismic zones in India.
Designing a structure in such a way that reducing damage during an earthquake makes the structure quite uneconomical, as the earthquake might or might not occurs in its life time and is a rare phenomenon. In this project G+6 existing Reinforced Concrete framed structure has been analyzed and designed using STAAD. PRO V8i. The main objective of this project is to compare the variation of steel percentage, maximum shear force, maximum bending moment, and maximum deflection in different seismic zones.

NARAYANA JUNIOR COLLEGE **Hyderabad, Telangana**
Marks – 842/1000 **June 2015 – April 2017**

Experience

CLOUDSWOOD CONSTRUCTIONS PVT LTD **Hyderabad, Telangana**
Civil Engineer **Dec 2023 – Present**

- Evaluated and interpreted testing and inspection data to assess the structural integrity of projects.
- Oversaw and inspected construction sites to ensure all work adhered to relevant building codes and regulations.
- Conducted comprehensive site visits and inspections to ensure compliance with safety standards and quality requirements.

Online Coursera Courses

3D CAD APPLICATIONS., National Taiwan University **June 2020 – July 2020**
Credential Link: https://www.coursera.org/account/accomplishments/verify/NZUFQJAPSGZS

MECHANICS OF MATERIALS 1: FUNDAMENTALS OF STRESS, STRAIN & AXIAL LOADING.,
Georgia Institute of Technology **October 2020 – November 2020**
Credential Link: https://www.coursera.org/account/accomplishments/verify/38QU26P6RLF9

BIM APPLICATIONS FOR ENGINEERS., National Taiwan University **October 2020 – December 2020**
Credential Link: https://www.coursera.org/account/accomplishments/verify/YX53GQB7ANWW

CONSTRUCTION COST ESTIMATING AND COST CONTROL.,
Columbia University, New York **November 2020 – December 2020**
Credential Link: https://www.coursera.org/account/accomplishments/verify/67FPUT96PL9T

ENGINEERING PROJECT MANAGEMENT: INITIATING AND PLANNING.,
Rice University **December 2020 – January 2021**
Credential Link: https://www.coursera.org/account/accomplishments/verify/4jR38MKVYBMK

Skills & Interests

Technical: Auto CAD (2D & 3D), Microsoft Office, Structural Analysis, Problem Solving, BIM (Structural & Architectural), Construction Management, Communication Skills, E-Tabs, Revit, Sketch UP.
Language: English, Hindi, Telugu.
Hobbies: Playing Cricket, Listening to Music.

Research Publications

1. **Advanced Geotechnical Simulations Using the MOOSE Framework: Techniques and Applications.**
    - This paper explores the use of the MOOSE (Multiphysics Object Oriented Simulation Environment) framework for geotechnical simulations, emphasizing its capabilities in transient, quasi-static processes. The study addresses model configuration, including strain and stress calculations, boundary fixation, and initial conditions such as gravitational and anisotropic stresses. Key advancements in geometry handling, such as subdomain-based management and mesh integration, are presented alongside strategies for simulating construction stages and material behaviors. The inclusion of external libraries like NEML2 and proven practices such as partial input file management and unit consistency ensures robust and efficient simulation setups. By leveraging MOOSE's massively parallel computing capabilities, this research aims to enhance geotechnical design methodologies and support innovative solutions in engineering practice.
2. **Advancing Soil Mechanics: A Two-Phase Contact MPM Algorithm for Saturated Soil-Structure Interaction.**
    - This study introduces a robust two-phase contact algorithm within the Material Point Method (MPM) framework to address these limitations. By modeling soil as a fully saturated porous medium and employing Chorin's projection method, the proposed approach reduces numerical oscillations and enables larger stable time steps. Additionally, an innovative rigid algorithm effectively manages the disparity in material stiffnesses between soil and single-phase materials. Comparative analyses with FEM and explicit MPM demonstrate the enhanced accuracy and efficiency of this method, offering a breakthrough in simulating hydro-mechanical interactions in soil-structure systems.
3. **Geotechnology Applications in Civil Engineering: Innovations in Sustainable Development and Infrastructure Management.**
    - This study explores the integration of geotechnology in civil engineering, emphasizing its applications in soil stabilization, geospatial mapping, and sustainable construction. The research highlights the use of advanced geotechnical software, remote sensing, and geographic information systems (GIS) to optimize infrastructure development. Key focus areas include earthquake-resistant design, landslide risk assessment, and foundation engineering. By leveraging innovative technologies, geotechnology not only improves the safety and efficiency of construction projects but also supports eco-friendly practices, aligning with the global push towards sustainable urbanization. This work provides insights into the transformative potential of geotechnology in addressing modern civil engineering challenges.